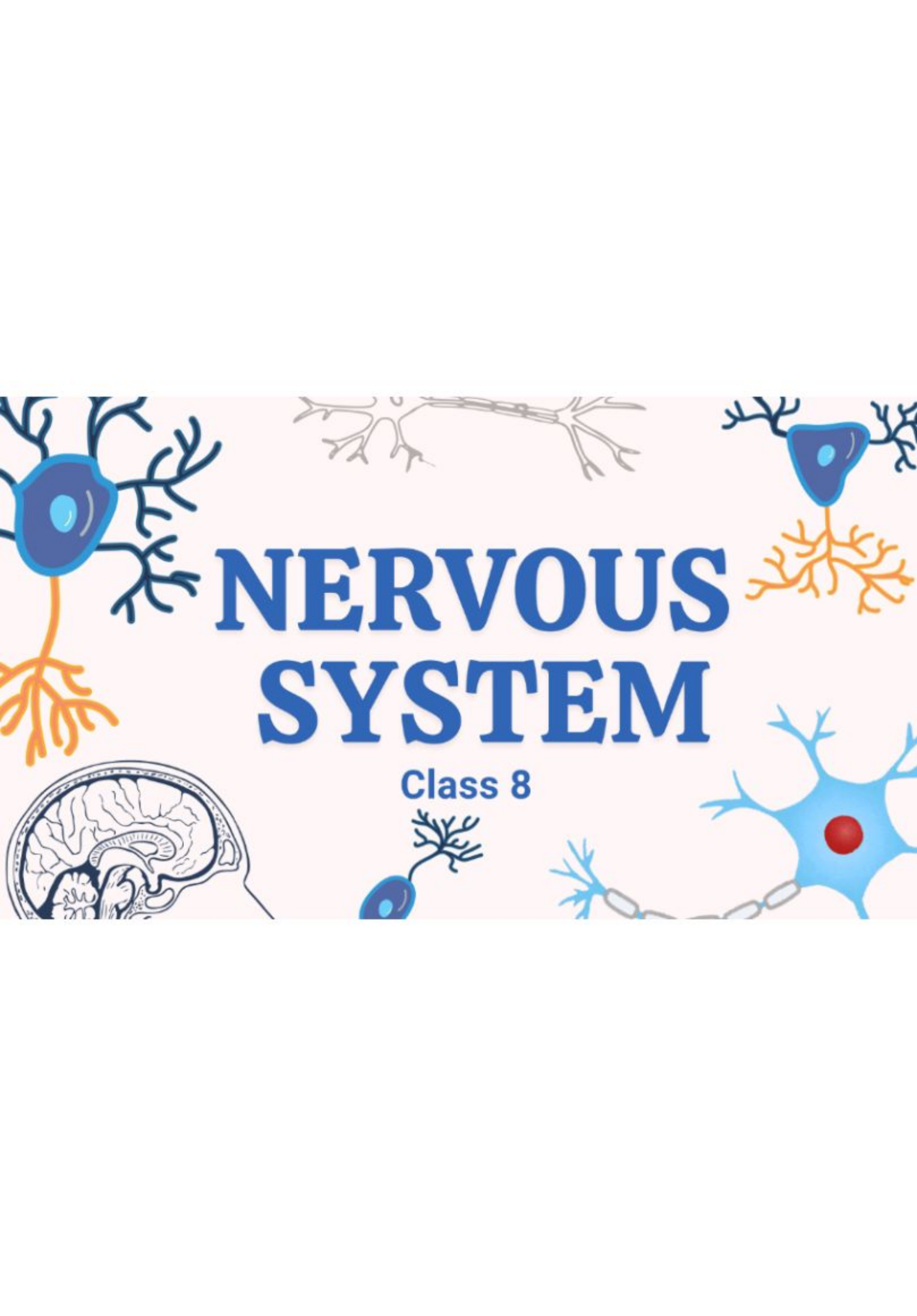
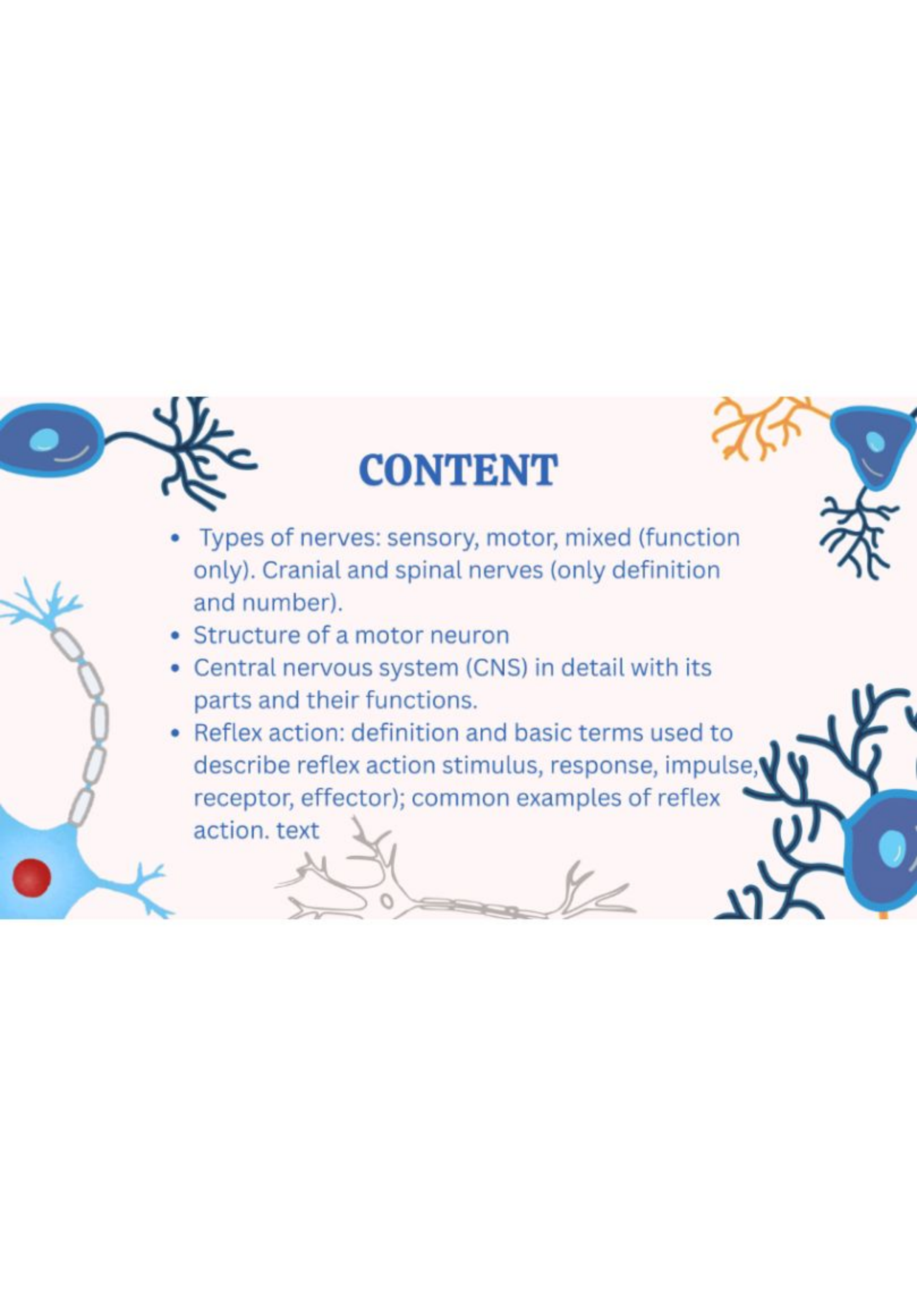




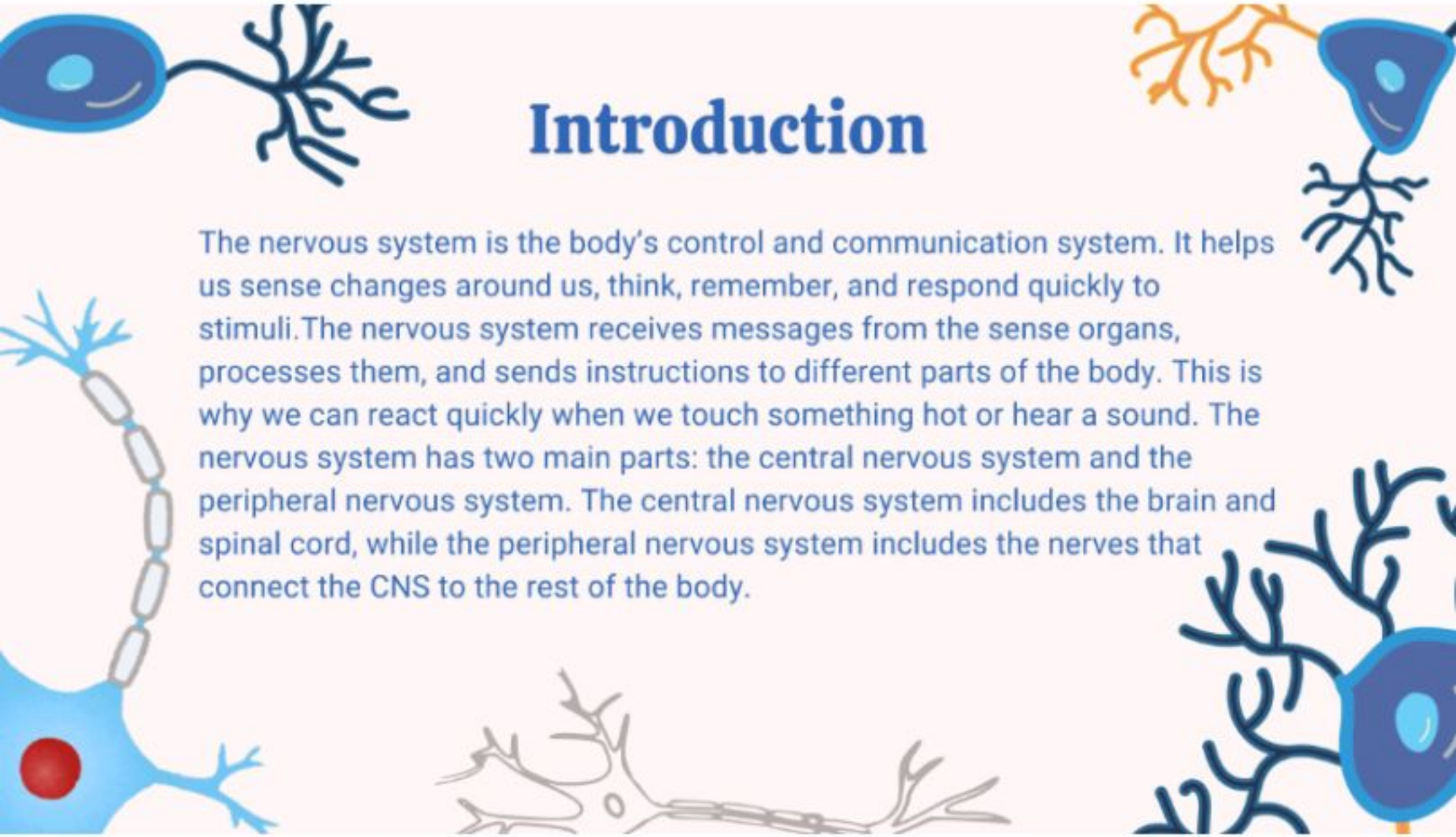
NERVOUS SYSTEM

Class 8



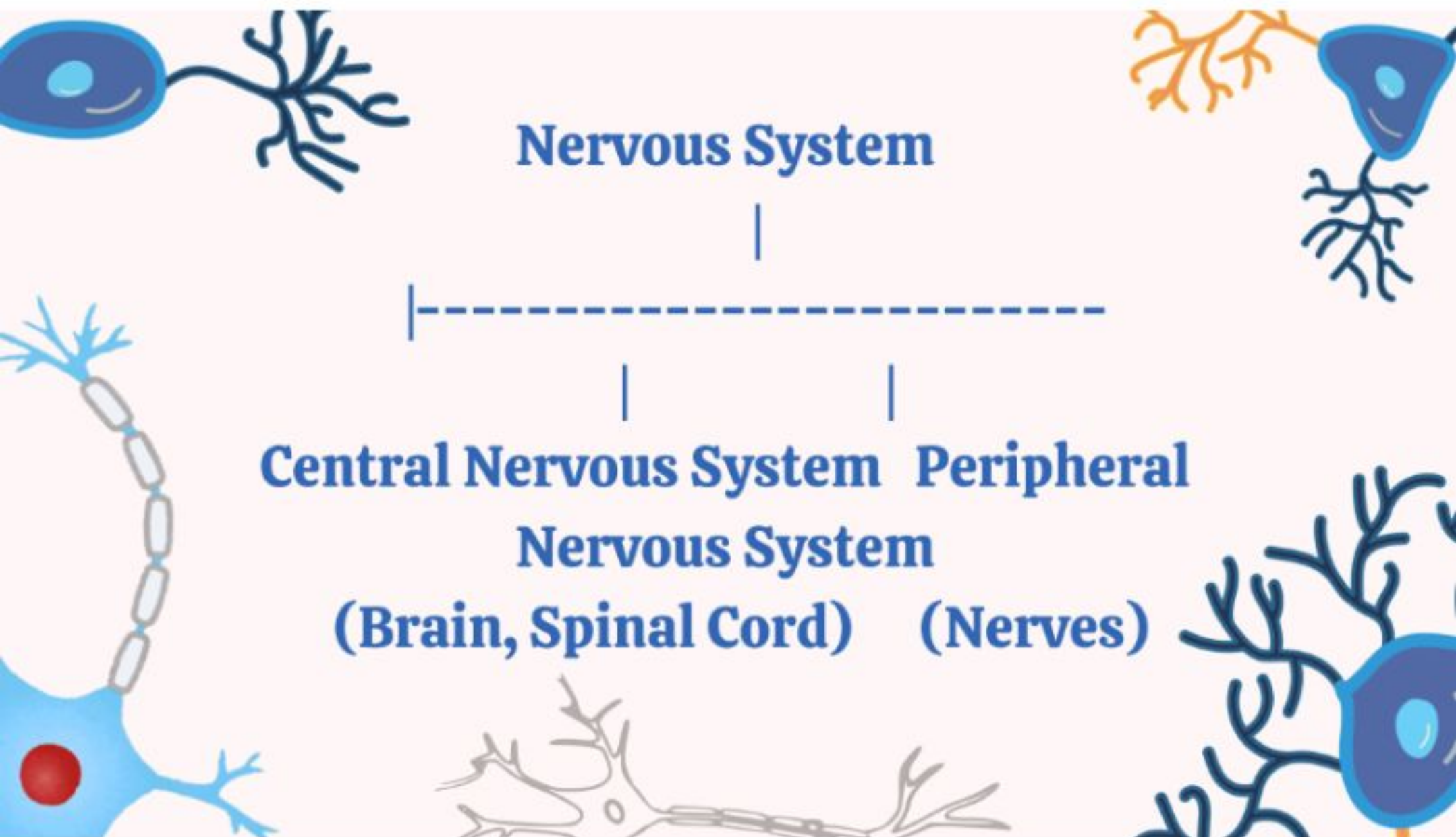
CONTENT

- Types of nerves: sensory, motor, mixed (function only). Cranial and spinal nerves (only definition and number).
- Structure of a motor neuron
- Central nervous system (CNS) in detail with its parts and their functions.
- Reflex action: definition and basic terms used to describe reflex action stimulus, response, impulse, receptor, effector); common examples of reflex action. text

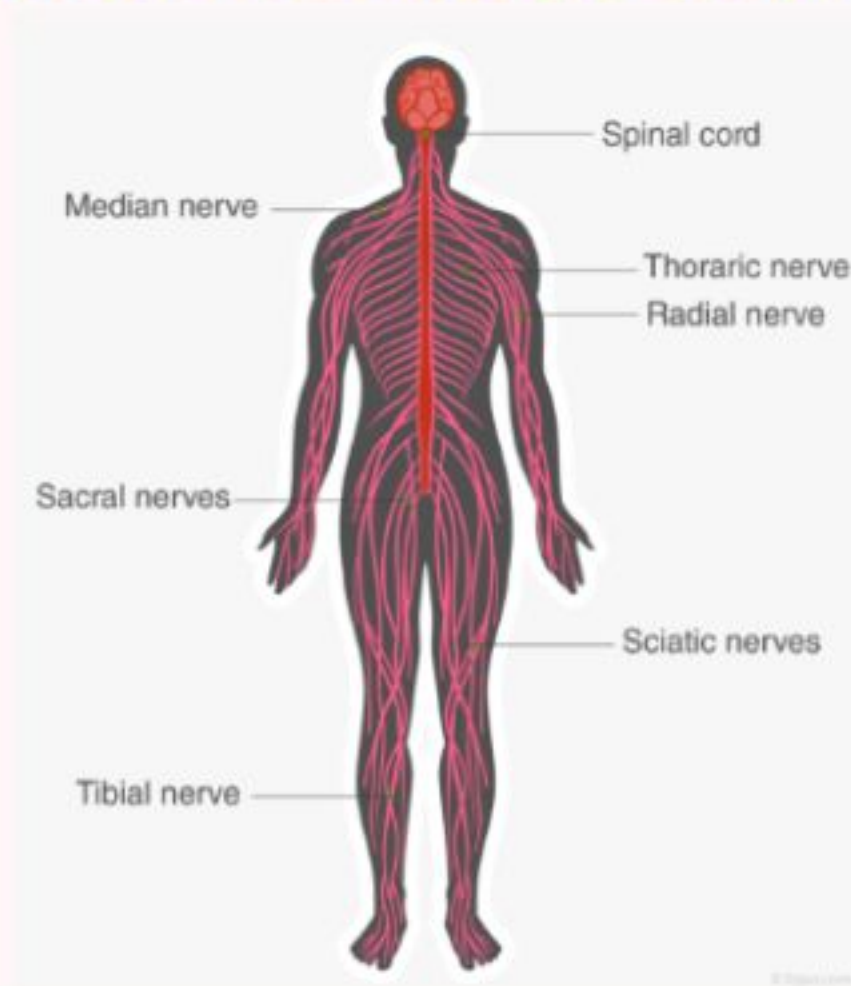


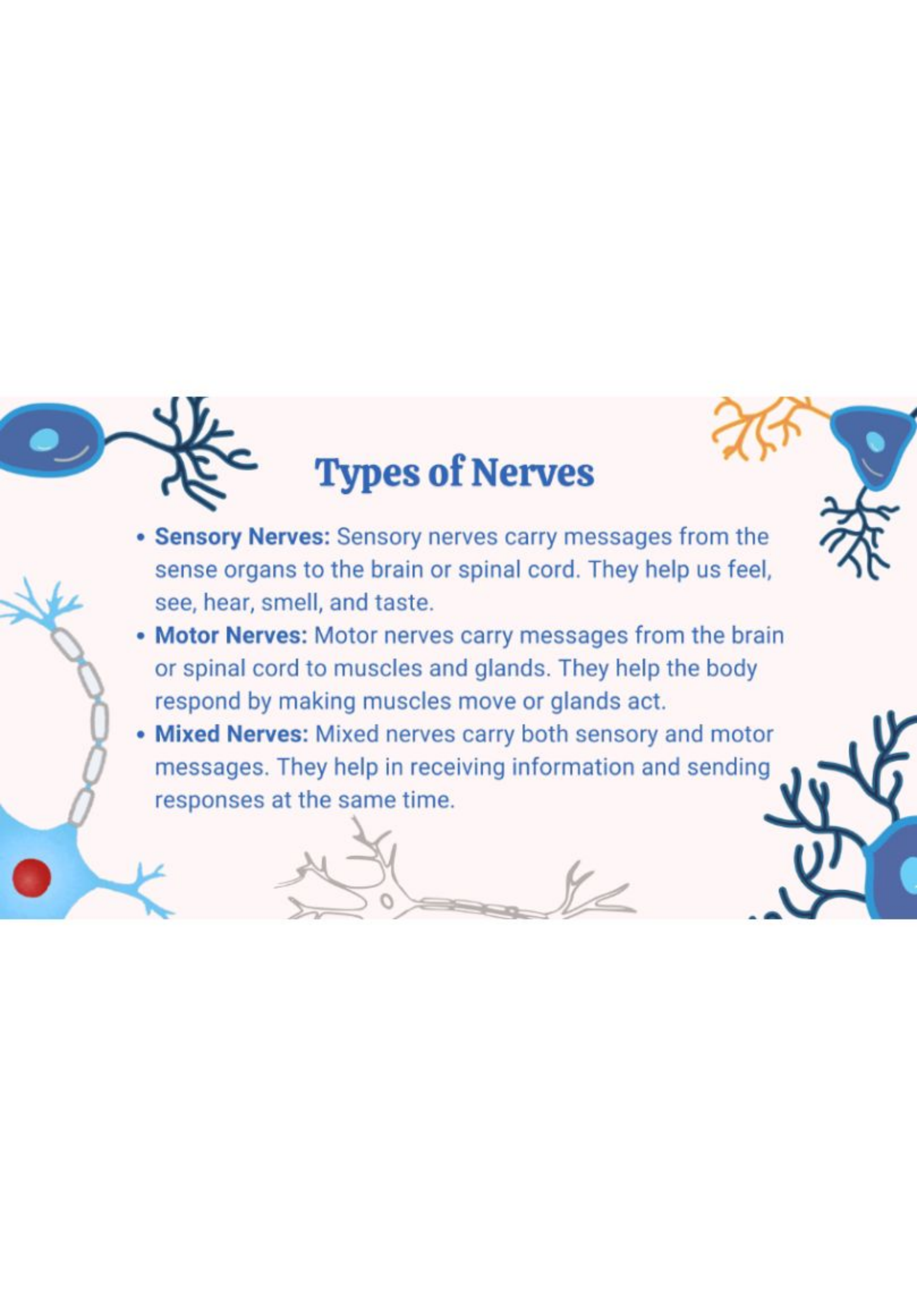
Introduction

The nervous system is the body's control and communication system. It helps us sense changes around us, think, remember, and respond quickly to stimuli. The nervous system receives messages from the sense organs, processes them, and sends instructions to different parts of the body. This is why we can react quickly when we touch something hot or hear a sound. The nervous system has two main parts: the central nervous system and the peripheral nervous system. The central nervous system includes the brain and spinal cord, while the peripheral nervous system includes the nerves that connect the CNS to the rest of the body.



HUMAN NERVOUS SYSTEM





Types of Nerves

- **Sensory Nerves:** Sensory nerves carry messages from the sense organs to the brain or spinal cord. They help us feel, see, hear, smell, and taste.
- **Motor Nerves:** Motor nerves carry messages from the brain or spinal cord to muscles and glands. They help the body respond by making muscles move or glands act.
- **Mixed Nerves:** Mixed nerves carry both sensory and motor messages. They help in receiving information and sending responses at the same time.



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SENSORY NEURONS

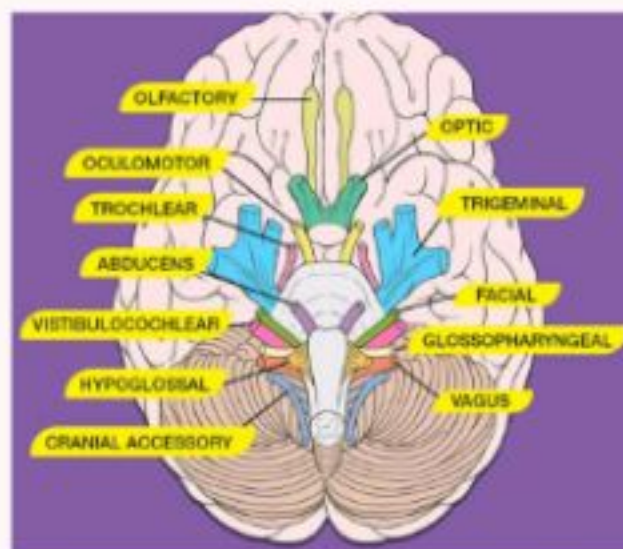
- Neurons that carry sensory impulse from sensory organs to the central nervous system are known as sensory neurons

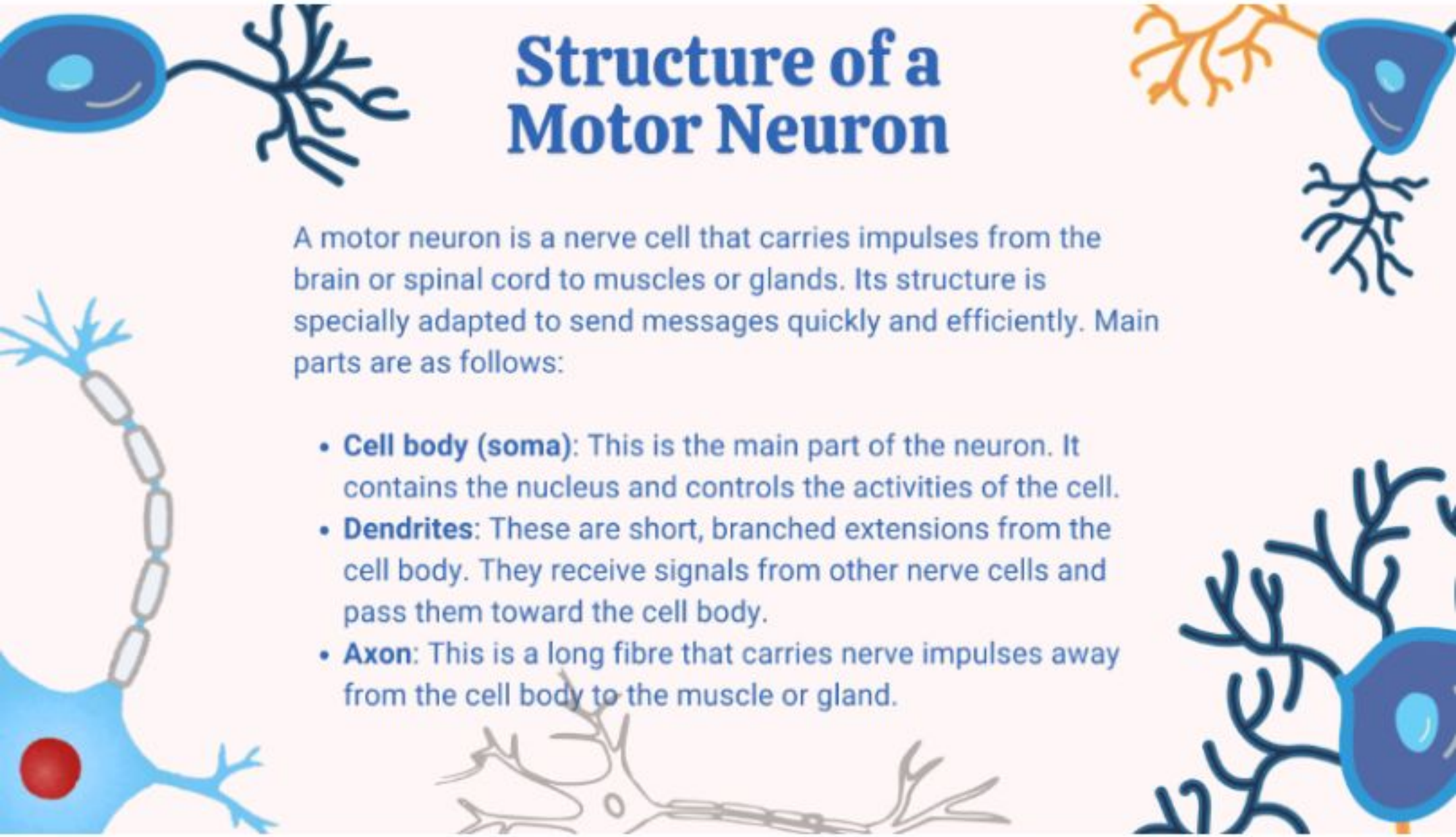
MOTOR NEURONS

- A neuron that carries motor impulses from the central nervous system to specific effectors is known as motor neurons.

Types of Nerves

- **Cranial Nerves:** Cranial nerves are the nerves that arise from the brain. There are 12 pairs of cranial nerves.
- **Spinal Nerves:** Spinal nerves are the nerves that arise from the spinal cord. There are 31 pairs of spinal nerves.

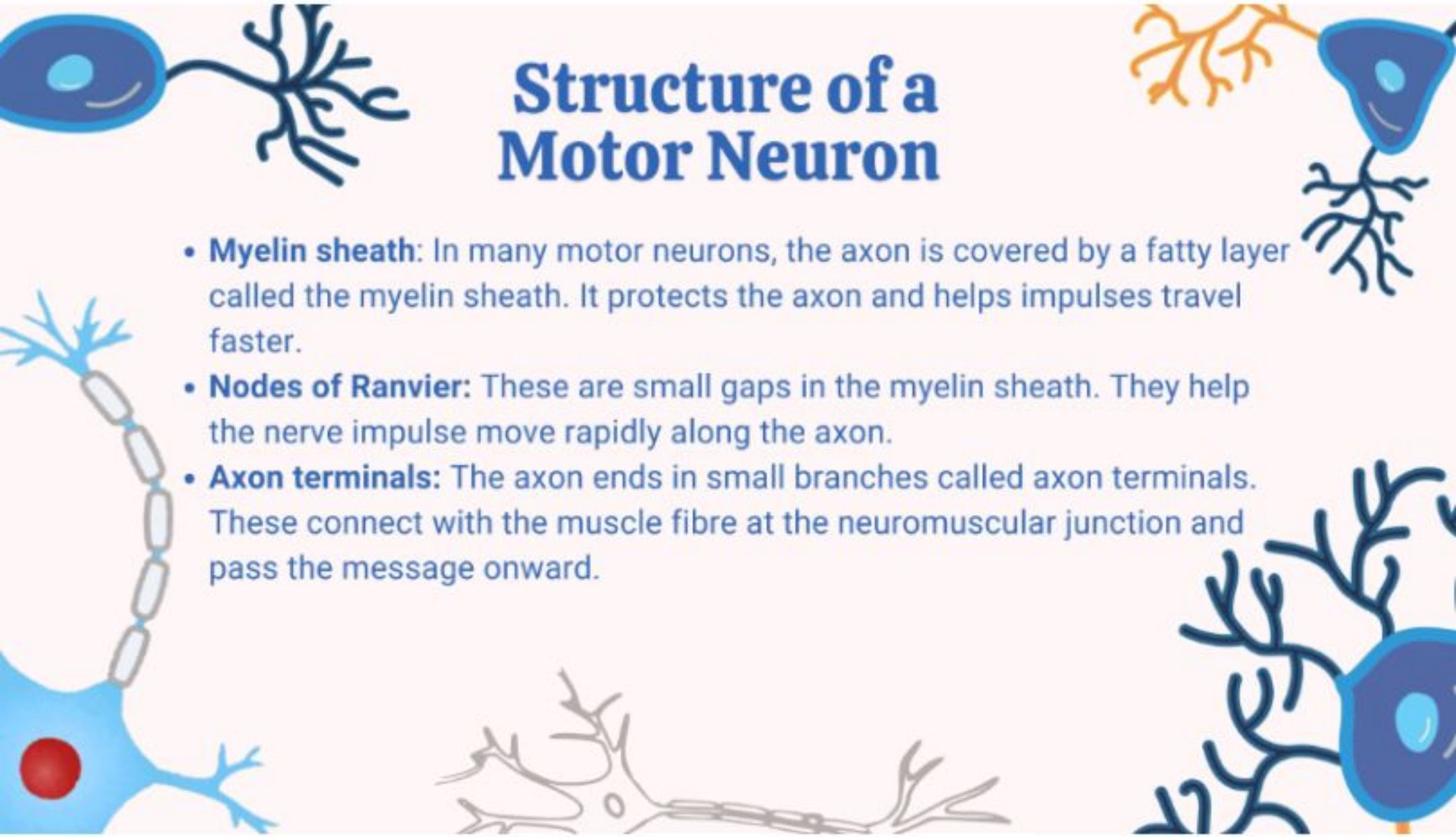


The page features several stylized illustrations of neurons. In the top left, a blue neuron with a large cell body and branching dendrites is shown. In the top right, a blue neuron with a triangular cell body and branching dendrites is depicted. In the bottom left, a blue neuron with a large cell body and a long axon is shown. In the bottom right, a blue neuron with a large cell body and branching dendrites is depicted. In the center, a blue neuron with a large cell body and a long axon is shown. The background is a light pink color.

Structure of a Motor Neuron

A motor neuron is a nerve cell that carries impulses from the brain or spinal cord to muscles or glands. Its structure is specially adapted to send messages quickly and efficiently. Main parts are as follows:

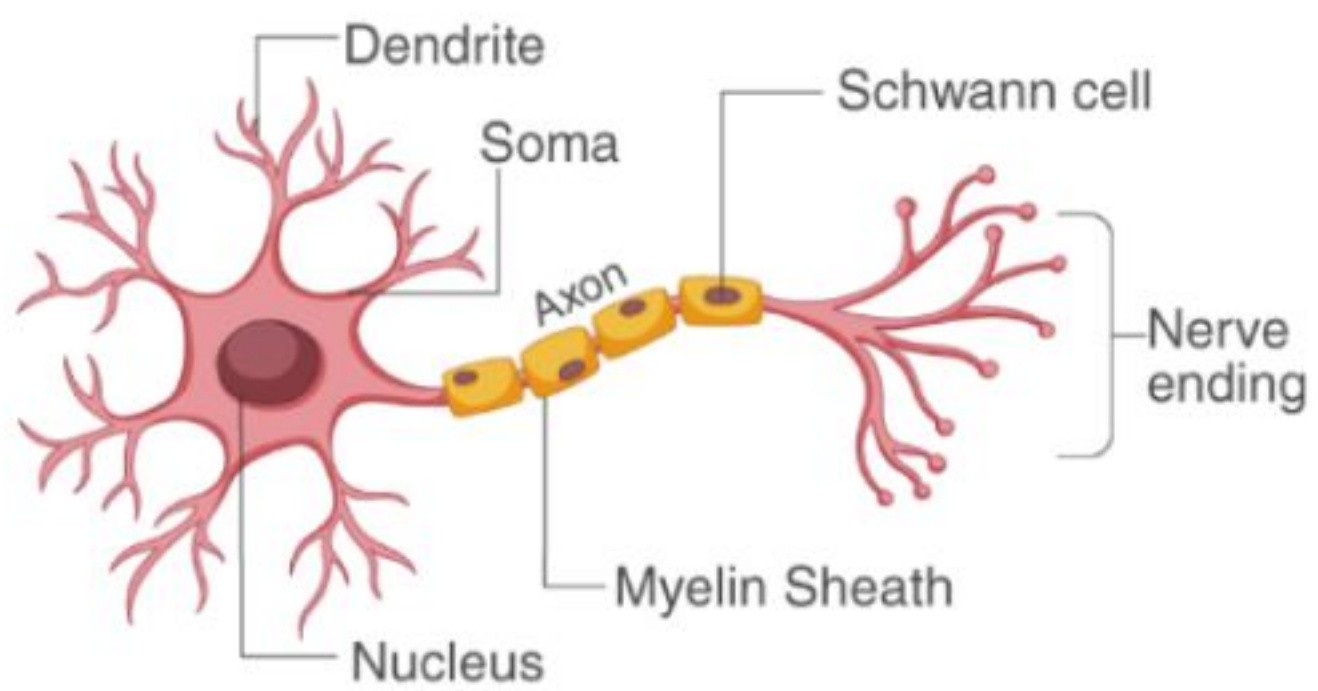
- **Cell body (soma):** This is the main part of the neuron. It contains the nucleus and controls the activities of the cell.
- **Dendrites:** These are short, branched extensions from the cell body. They receive signals from other nerve cells and pass them toward the cell body.
- **Axon:** This is a long fibre that carries nerve impulses away from the cell body to the muscle or gland.

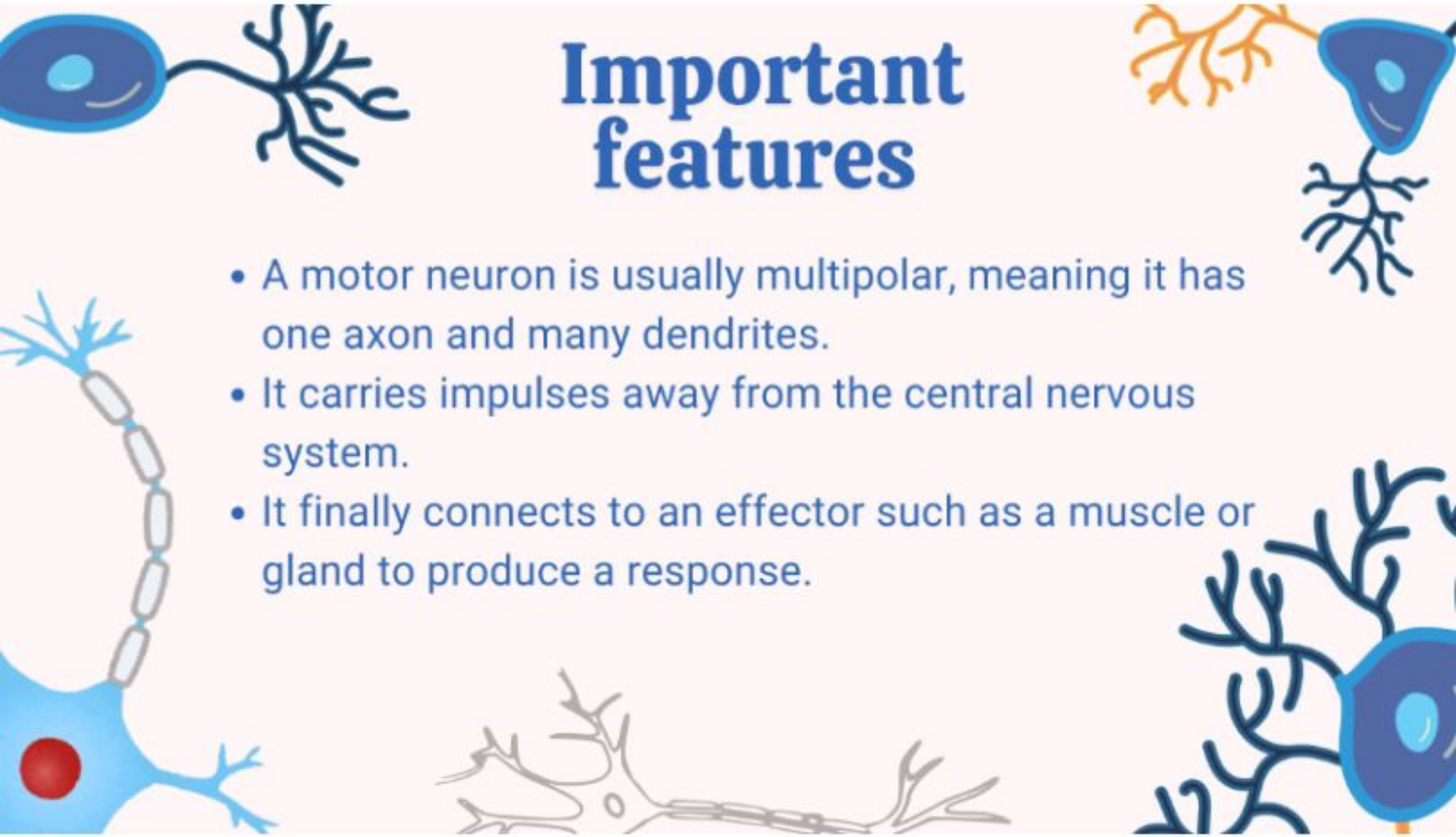
The background of the slide features several stylized illustrations of neurons. In the top left, a blue neuron with a large cell body and branching dendrites is shown. In the top right, an orange neuron with a triangular cell body and branching dendrites is depicted. On the left side, a blue neuron with a red nucleus and a long axon covered in a myelin sheath is shown. On the right side, a blue neuron with a large cell body and branching dendrites is shown. At the bottom center, a grey neuron with a long axon and branching terminals is depicted.

Structure of a Motor Neuron

- **Myelin sheath:** In many motor neurons, the axon is covered by a fatty layer called the myelin sheath. It protects the axon and helps impulses travel faster.
- **Nodes of Ranvier:** These are small gaps in the myelin sheath. They help the nerve impulse move rapidly along the axon.
- **Axon terminals:** The axon ends in small branches called axon terminals. These connect with the muscle fibre at the neuromuscular junction and pass the message onward.

STRUCTURE OF NEURON



The slide features several stylized illustrations of neurons. In the top left, a blue neuron with a large nucleus and branching dendrites. In the top right, an orange neuron with a large nucleus and branching dendrites. In the bottom left, a blue neuron with a large nucleus and a long axon covered in myelin sheaths. In the bottom right, a blue neuron with a large nucleus and branching dendrites. In the bottom center, a grey neuron with a large nucleus and branching dendrites.

Important features

- A motor neuron is usually multipolar, meaning it has one axon and many dendrites.
- It carries impulses away from the central nervous system.
- It finally connects to an effector such as a muscle or gland to produce a response.

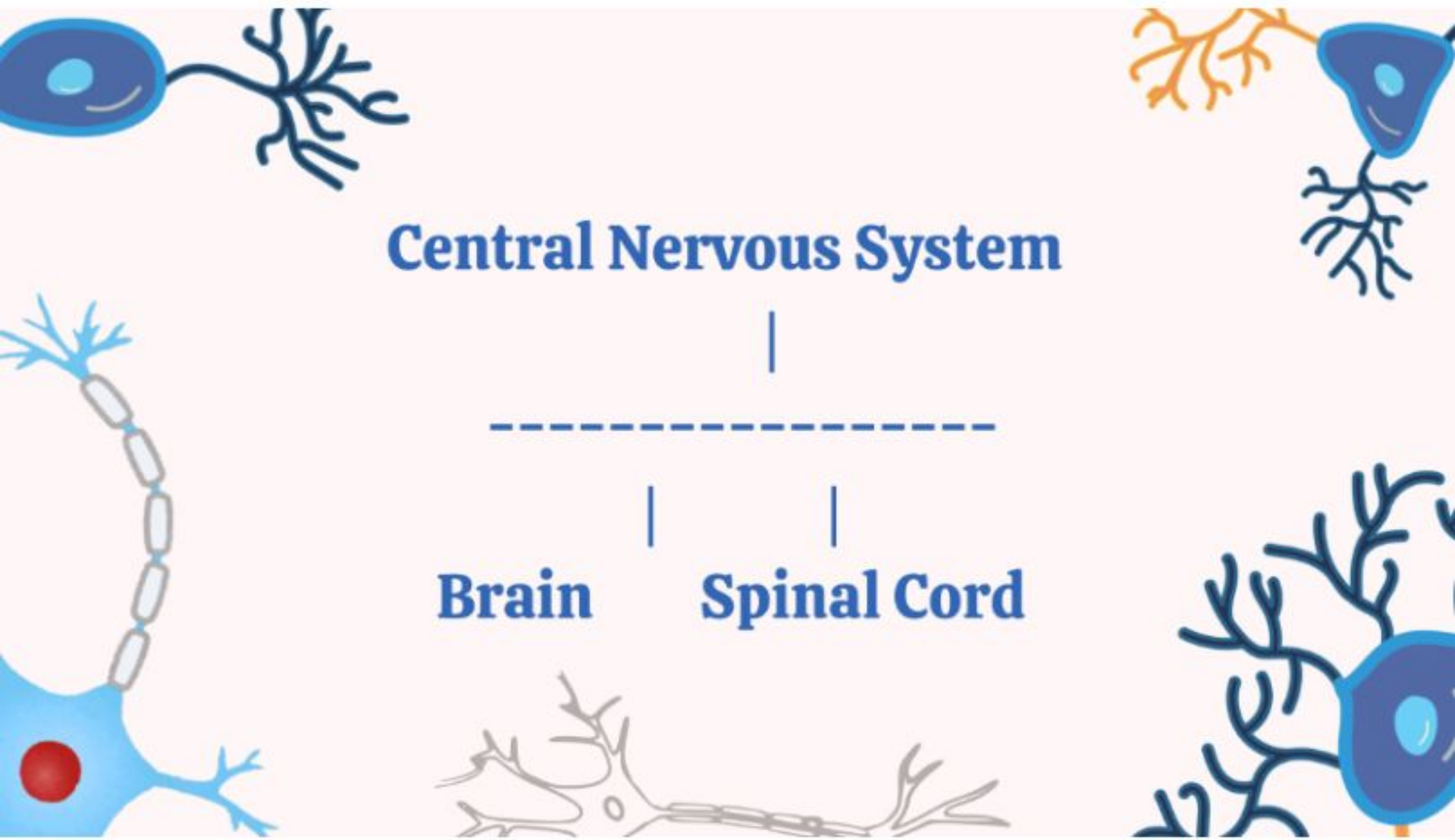


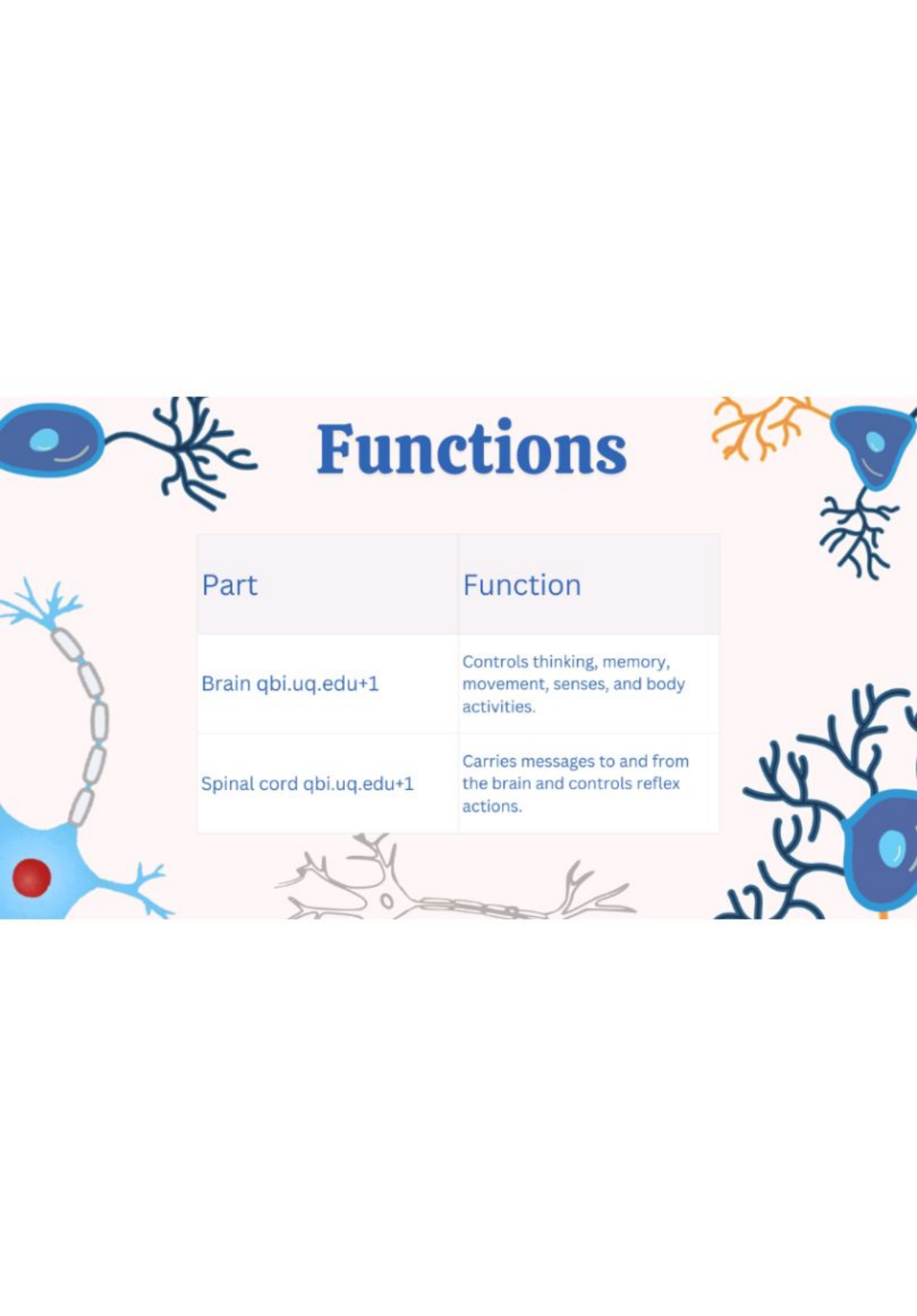
Central Nervous System



The central nervous system is the main control and processing center of the body. It is made up of the brain and spinal cord. Parts of the CNS are as follows:

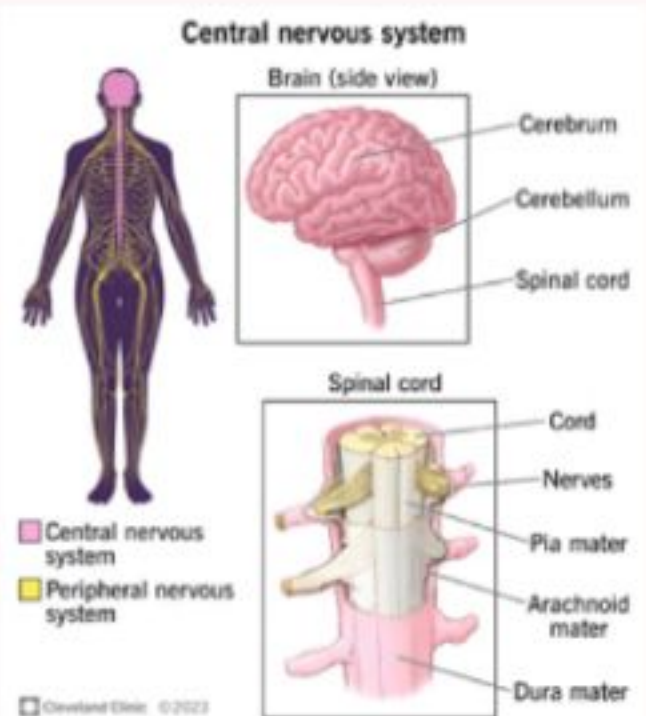
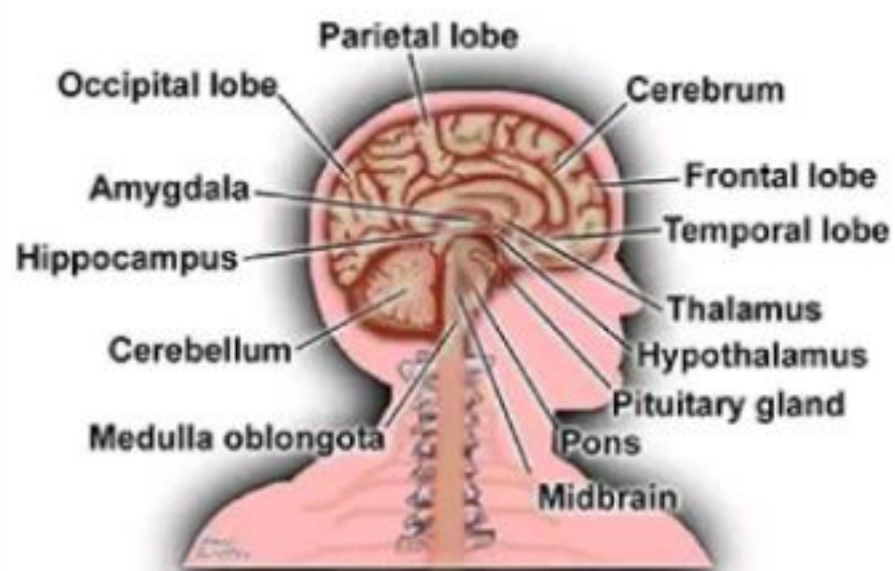
- **BRAIN:** The brain is the control centre of the body. It receives information, thinks, remembers, and sends instructions to different body parts.
 - **SPINAL CORD:** The spinal cord is a long, cord-like structure that connects the brain to the rest of the body. It carries messages between the brain and spinal nerves and also helps in reflex actions.
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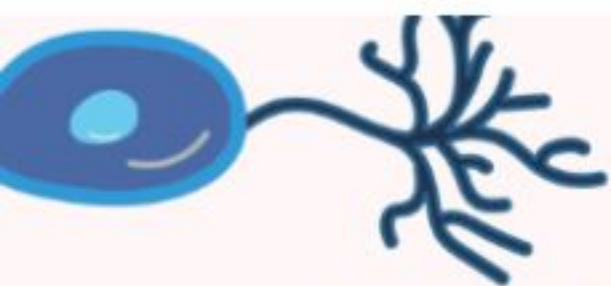




Functions

Part	Function
Brain qbi.uq.edu+1	Controls thinking, memory, movement, senses, and body activities.
Spinal cord qbi.uq.edu+1	Carries messages to and from the brain and controls reflex actions.





Reflex action

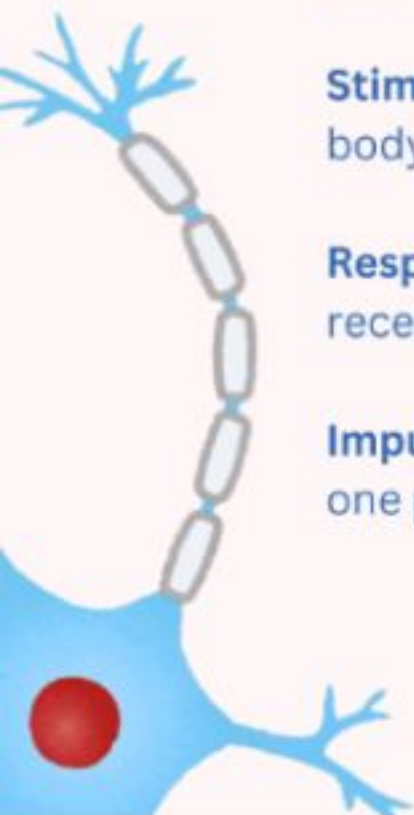


A reflex action is a quick, automatic, and involuntary response to a stimulus. It usually helps protect the body from harm.

Stimulus: A stimulus is any change in the surroundings or inside the body that causes a response, such as heat, light, touch, or pain.

Response: A response is the action or reaction shown by the body after receiving a stimulus, such as pulling the hand away from a hot object.

Impuls: An impulse is a nerve message carried through nerve cells from one part of the body to another.





Reflex action

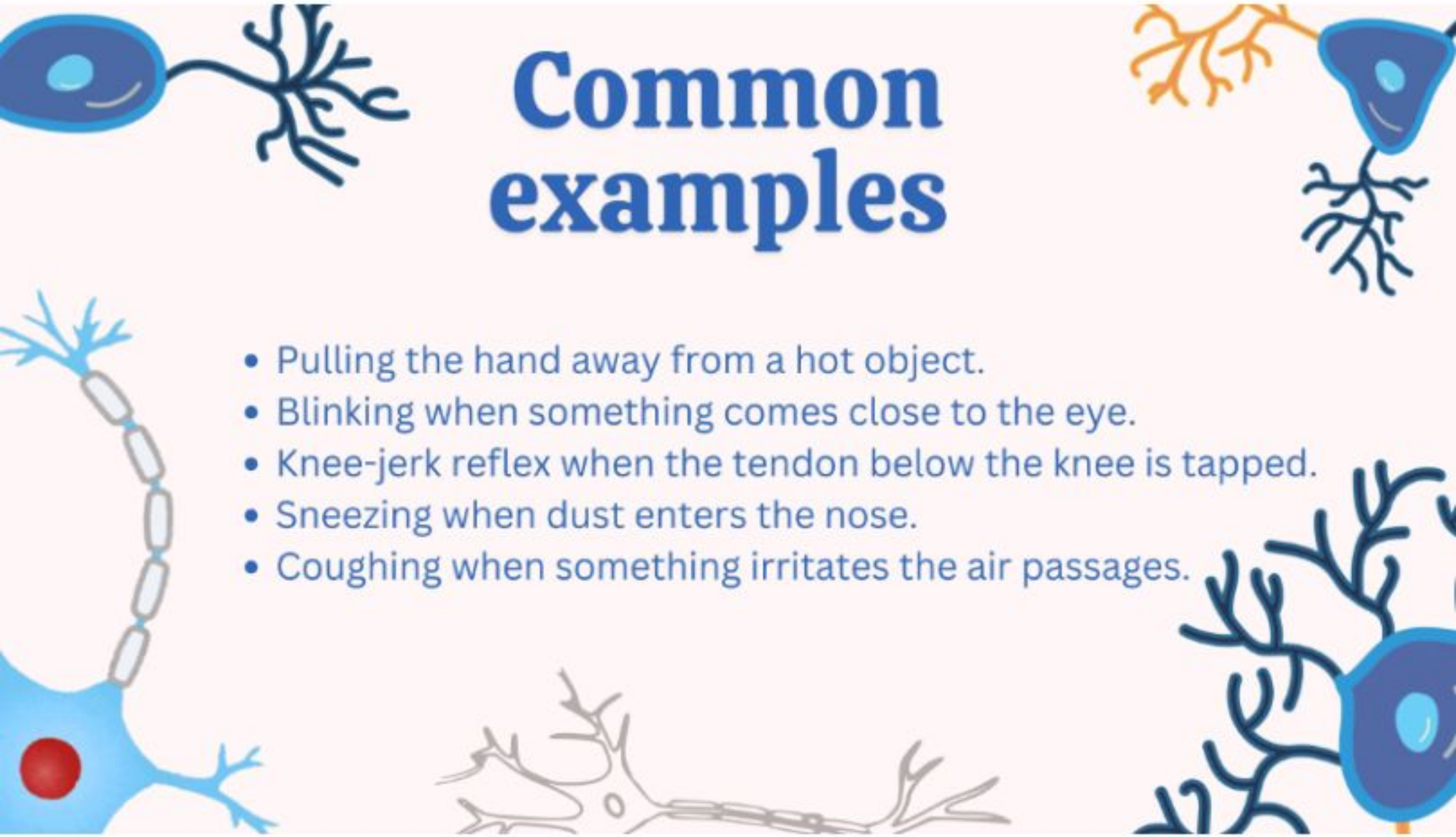


Receptor: A receptor is a sense cell or nerve ending that detects the stimulus and starts the nerve message.



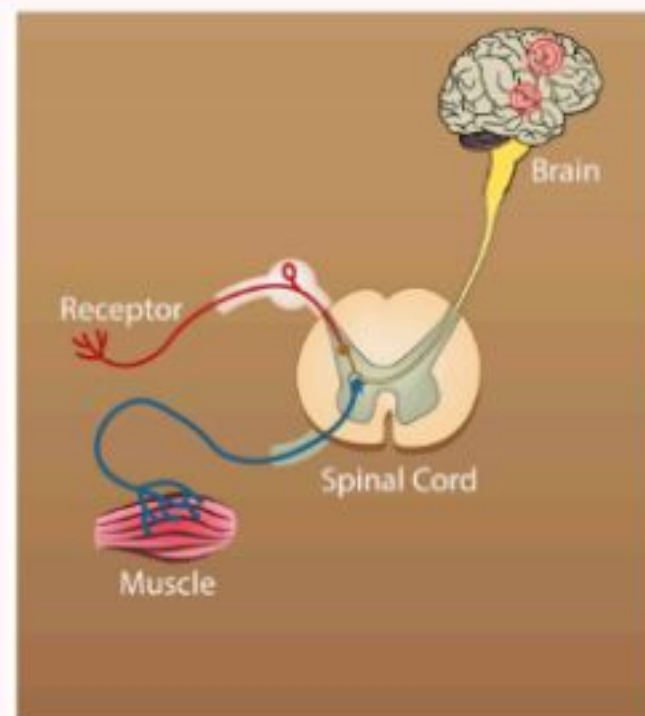
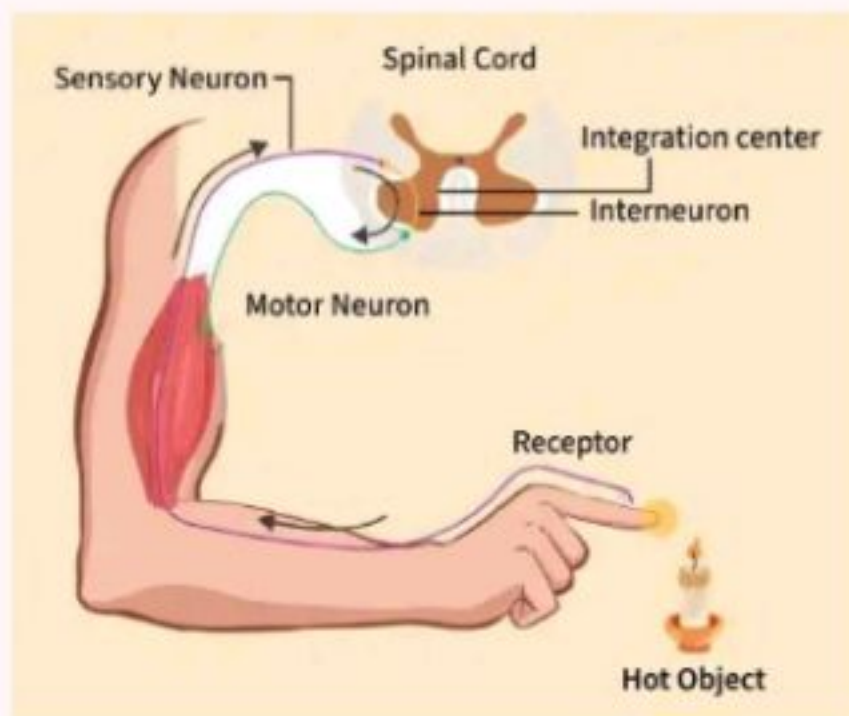
Effector: An effector is a muscle or gland that carries out the response, such as a muscle contracting to withdraw the hand



The slide features several stylized neuron illustrations. In the top left, a blue neuron with a circular cell body and branching dendrites. In the top right, a blue neuron with a triangular cell body and branching dendrites, with orange branching structures above it. In the bottom left, a blue neuron with a circular cell body containing a red nucleus and a long, segmented axon. In the bottom right, a blue neuron with a circular cell body and branching dendrites. In the bottom center, a grey, branching structure resembling a neuron or a small tree.

Common examples

- Pulling the hand away from a hot object.
- Blinking when something comes close to the eye.
- Knee-jerk reflex when the tendon below the knee is tapped.
- Sneezing when dust enters the nose.
- Coughing when something irritates the air passages.

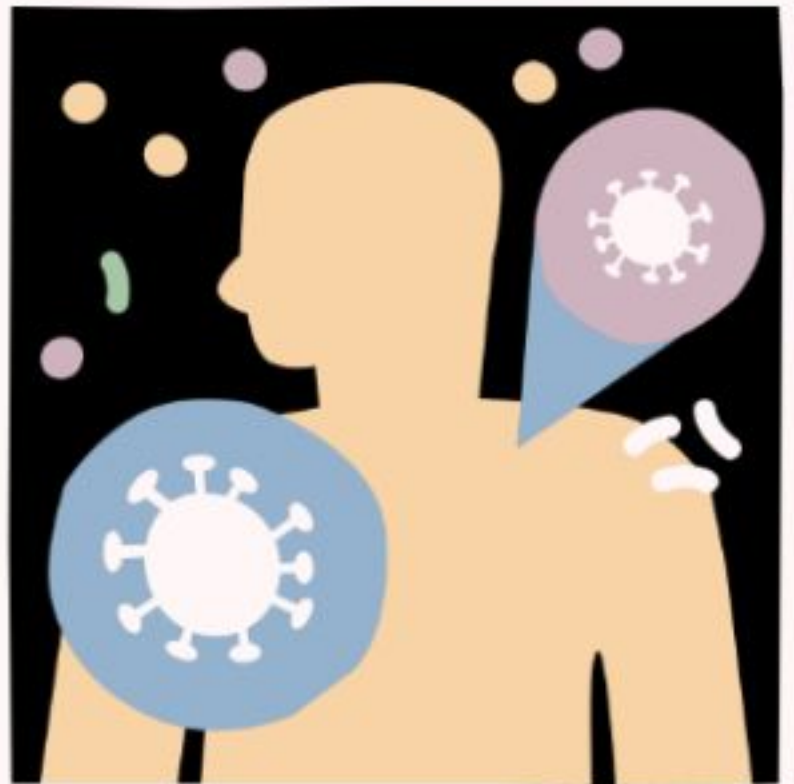


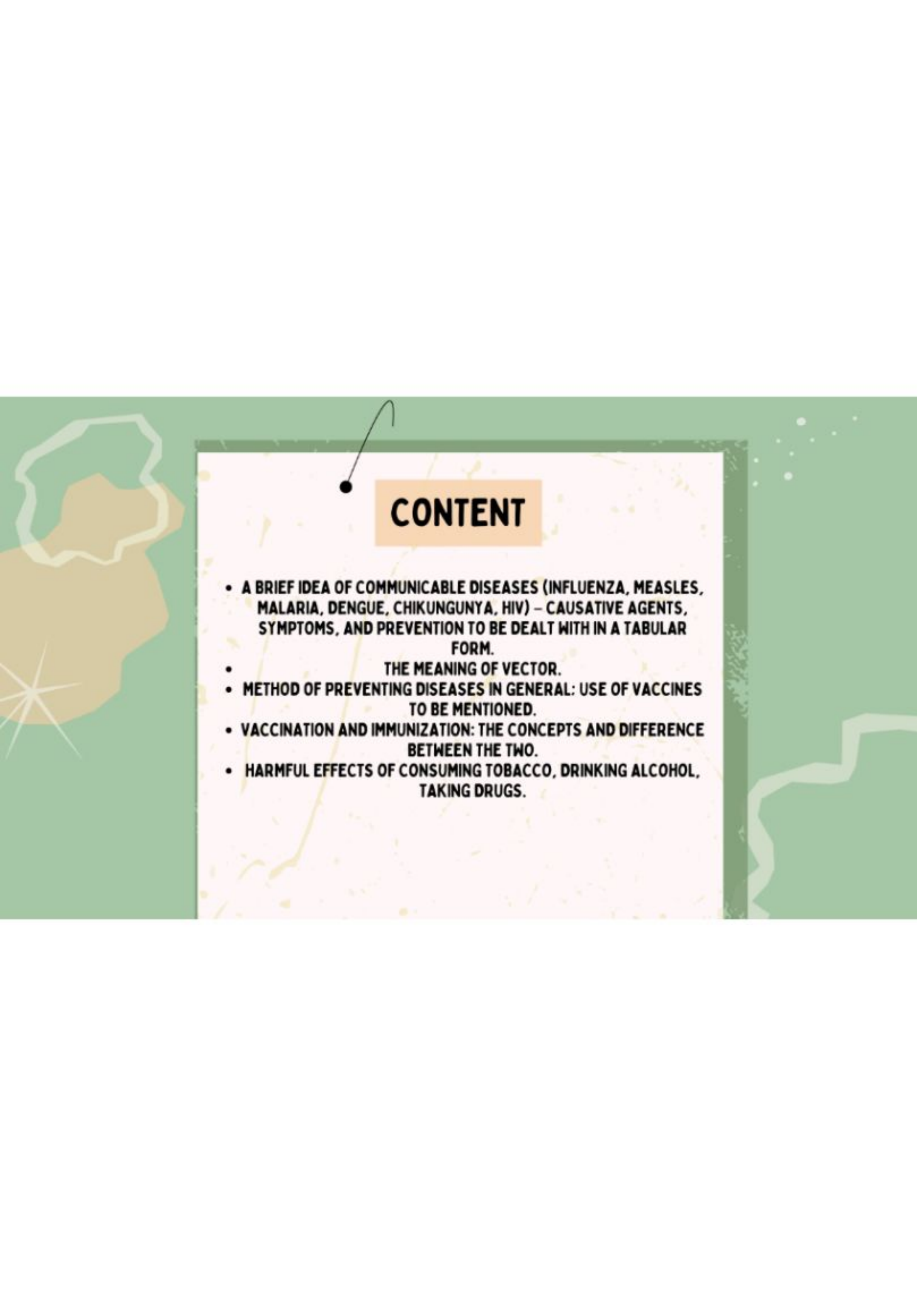
END



HEALTH & HYGIENE DISEASES

CLASS 8





CONTENT

- **A BRIEF IDEA OF COMMUNICABLE DISEASES (INFLUENZA, MEASLES, MALARIA, DENGUE, CHIKUNGUNYA, HIV) – CAUSATIVE AGENTS, SYMPTOMS, AND PREVENTION TO BE DEALT WITH IN A TABULAR FORM.**
- **THE MEANING OF VECTOR.**
- **METHOD OF PREVENTING DISEASES IN GENERAL: USE OF VACCINES TO BE MENTIONED.**
- **VACCINATION AND IMMUNIZATION: THE CONCEPTS AND DIFFERENCE BETWEEN THE TWO.**
- **HARMFUL EFFECTS OF CONSUMING TOBACCO, DRINKING ALCOHOL, TAKING DRUGS.**

INTRODUCTION

Has this ever happened to you? A student next to you in class has been coughing and sneezing all day. You feel fine that day, but you also come down with a cold two days later.

What do you think is the reason behind this?





COMMUNICABLE DISEASE

Communicable diseases are diseases that spread from one person to another, or from animals, insects, contaminated food, water, or air. They are caused by disease-causing microorganisms such as viruses, bacteria, parasites, or protozoa.



COMMUNICABLE DISEASE

Disease	Causative agent	Main symptoms	Prevention
Influenza	Virus	Fever, cough, sore throat, body pain, weakness	Personal hygiene, covering mouth and nose while coughing, vaccination
Measles	Virus	Fever, rash, cough, red eyes, runny nose	Vaccination, isolation of infected person, cleanliness
Malaria	Protozoa (Plasmodium)	Fever with chills, sweating, headache, weakness	Avoid mosquito bites, use nets, remove stagnant water
Dengue	Virus	High fever, headache, pain in joints and muscles, rash	Prevent mosquito breeding, use mosquito nets and repellents
Chikungunya	Virus	Fever, severe joint pain, headache, rash	Mosquito control, protective clothing, remove stagnant water
HIV/AIDS	Virus	Weak immunity, repeated infections, weight loss, tiredness	Safe blood transfusion, safe sex, awareness, avoid sharing needles



WHAT IS VECTOR



A vector is an organism, usually an insect, that carries disease-causing germs from one person, animal, or place to another without getting the disease itself. Mosquitoes are common vectors for malaria, dengue, and chikungunya.



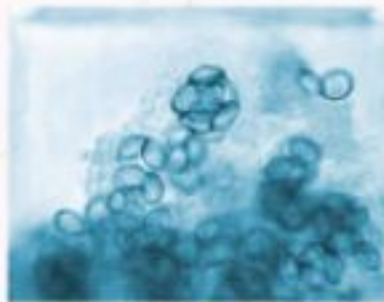
PATHOGENS

These are microorganisms that **cause diseases**.
A disease caused by a pathogen is called a **communicable disease**,
meaning it can spread from one organism to another organism.

FOUR MAIN TYPES OF PATHOGENS:



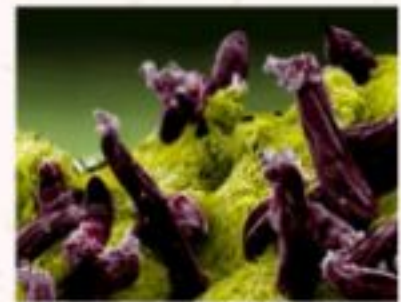
Bacteria



Fungi



Virus



Protist

HOW DO PATHOGENS SPREAD?

Direct Contact

Involves physical interaction with infected individuals, such as through blood or bodily fluids



Indirect Contact

Involves exposure to pathogens via contaminated surfaces, food, soil or air



Vector

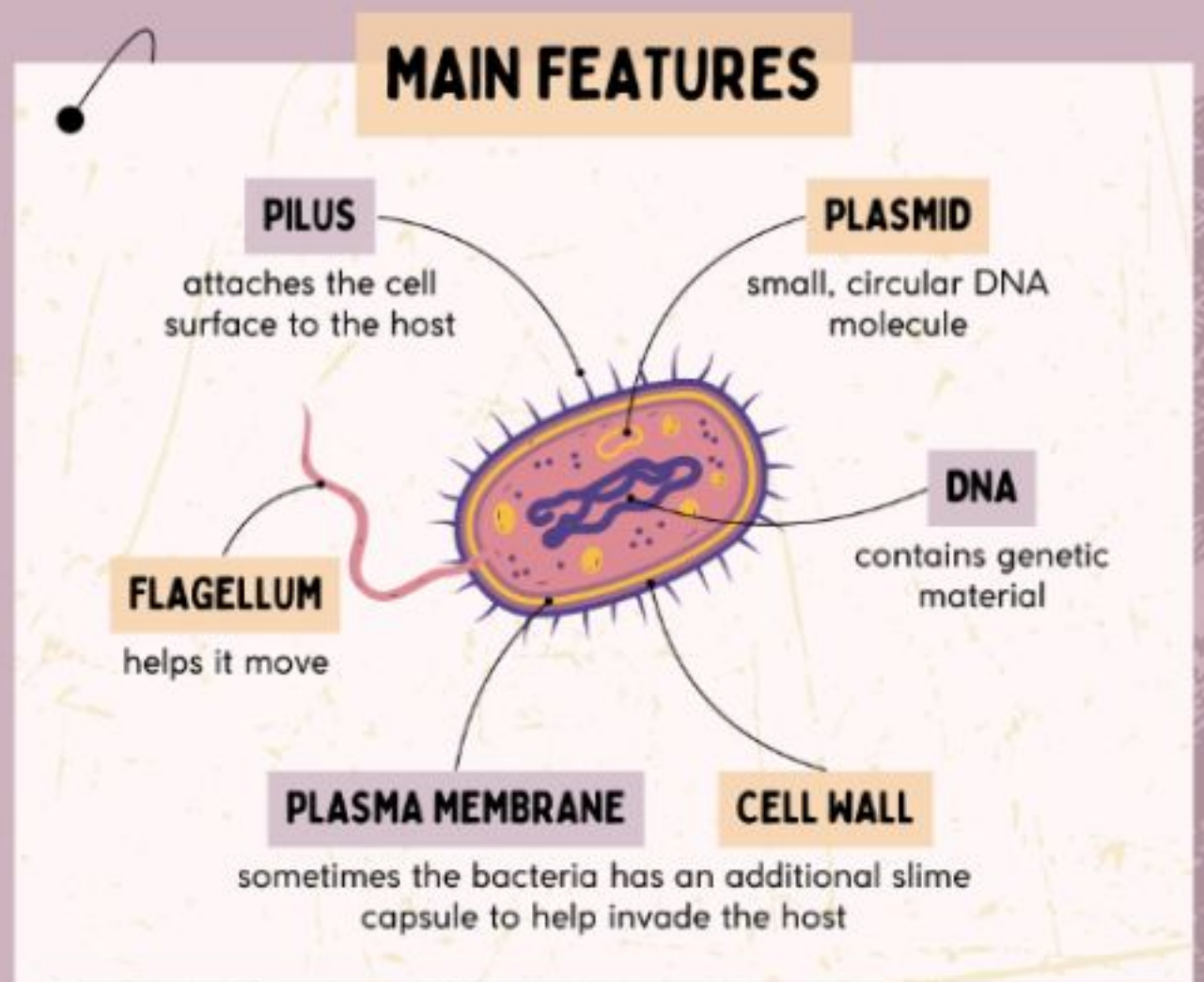
Requires organisms to spread diseases, such as mosquitos, mice and ticks



BACTERIA

These are **single-celled organisms** that lack a nucleus.

Even though bacterial cells have a simple structure, they vary in different ways depending on the species.



DISEASES CAUSED BY BACTERIA



***Salmonella* Infection**

It is caused by the bacteria *Salmonella*, which is present in contaminated foods and surfaces. Symptoms include abdominal pain, vomiting, and diarrhoea.



Citrus Canker

It is caused by the bacterium *Xanthomonas*. It manifests as lesions with a yellow halo affecting stems, leaves, and citrus fruits.

PREVENTION OF DISEASES IN GENERAL

Diseases can be prevented by maintaining good hygiene, eating clean and nutritious food, drinking safe water, keeping the surroundings clean, and avoiding contact with infected persons. Vaccines also help prevent many diseases by preparing the body to fight germs before illness starts. Use of Vaccines

A vaccine is a substance given to a person to protect against a specific disease. Vaccines help the body produce protection against germs and reduce the chance of getting infected.

VACCINATION AND IMMUNIZATION

Vaccination means giving a vaccine to a person.

Immunization means becoming protected against a disease after vaccination or exposure to a disease.

Vaccination	Immunization
It is the act of giving a vaccine.	It is the result of protection developed in the body.
It is a process.	It is the body's immunity against a disease.
Done by a doctor or health worker.	Happens after the vaccine works in the body.

HARMFUL EFFECTS OF TOBACCO, ALCOHOL, AND DRUGS

- **TOBACCO:** Tobacco use is harmful because it damages the lungs, causes cough and breathing problems, and may lead to diseases like cancer and heart problems. It also harms people who breathe the smoke of others.
- **ALCOHOL:** Drinking alcohol is harmful because it affects the brain, weakens judgement, and damages the liver and other organs. It may also cause addiction and poor health.
- **DRUGS:** Taking drugs without medical need is dangerous because it can damage the brain and body, cause addiction, and disturb normal life. Drug abuse may lead to serious illness and social problems.

